

Life as Boundary-Condition Updateability: Topological Connectivity as a Candidate Physical Signature of the Living State

A Companion Framework to the Volume Phase Transition Hypothesis of Chronic Disease

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Abstract

We recently proposed that the volume phase transition (VPT) of interstitial polyanionic gel constitutes the mesoscale physical mechanism underlying Prigogine's internal time, and that chronic disease represents a thermodynamic state in which gel collapse has attenuated a system's capacity for dissipative self-organization. Here we generalize this account into a more fundamental physical criterion for the living state itself: boundary-condition updateability. We define life as a physical condition in which the junction between past structure and future possibility remains open through the continuous, internal modifiability of the boundary conditions governing energy flow, transport pathways, and dissipative degrees of freedom. We argue that this updateability is supported by nontrivial topological connectivity within the organism-wide mesoscopic transport field -- the interstitial network -- and propose the first Betti number ($b_1 \geq 1$) as one candidate, measurable structural signature of such connectivity, ensuring continuous loops and alternative pathways through which entropy can be distributed rather than locally trapped. We distinguish geometric connectivity, which describes anatomical arrangement, from effective connectivity, which describes the transport actually available; gel collapse can reduce the latter toward zero while leaving the former intact, which may explain why the collapsed state resists detection by conventional structural imaging. This account converges with, and extends, Igamberdiev's recent re-evaluation of Bauer's principle of sustained non-equilibrium, situating topological connectivity as a candidate structural precondition through which thermodynamic buffering becomes architecturally possible. We further note that recent anatomical work has independently demonstrated the body-wide continuity of interstitial spaces -- across organ boundaries and along the adventitia of the vasculature and the perineurium of nerves -- providing empirical support for a connected transport field consistent with this framework. On this account the interstitium is not a specialized tissue that happens to matter, but the particular form taken, in multicellular organisms, by a transport field that all life requires: in unicellular organisms the external medium itself, at the origin of life the ocean, serves the same function. We formulate this criterion in

terms of Markovian dynamics: living systems are characteristically history-dependent, carrying their history forward materially while keeping their reachable future state space open, and may therefore require non-Markovian description at appropriate coarse-grained scales; the loss of updateability is accordingly a process of Markovianization, in which history-dependent degrees of freedom are progressively erased and the accessible future state space contracts. This account converges with, while extending, the theory of autopoiesis, to which it proposes to add a candidate physical substrate -- the interstitial phase state -- for the self-production that autopoiesis specifies organizationally but, as acknowledged within that literature, does not ground in thermodynamic terms. Death, on this account, is not defined by the cessation of energy flow alone, but by the loss of effective capacity to update the boundary conditions governing that flow; a collapse of functionally relevant topological connectivity may constitute one physical route by which this loss occurs.

Keywords: *boundary-condition updateability; non-Markovian dynamics; autopoiesis; topological connectivity; Betti number; interstitium; non-equilibrium thermodynamics; volume phase transition; sustained non-equilibrium; Bauer's principle*

1. Introduction: From Gel Collapse to a General Physical Criterion for Life

In a companion paper, we proposed that the interstitial matrix of multicellular organisms undergoes volume phase transitions (VPT) between a swollen and a collapsed phase, and that this transition constitutes the mesoscale physical mechanism by which thermodynamic closure occurs in living tissue -- a mechanism operating at a scale that Prigogine's framework of internal time and dissipative structures did not address [1]. That account was developed specifically around the interstitial gel and its clinical consequences for chronic disease.

The present essay generalizes the underlying physical intuition into a broader criterion, applicable in principle beyond the interstitial gel of the specific case: life as the continuous updateability of internal boundary conditions. Where the companion paper asked what physical state of the interstitium permits or forecloses dissipative self-organization, this essay asks a prior question: what general physical property must any open system possess in order to remain, in a meaningful sense, alive -- capable of converting the structure inherited from its past into coherent activity in its future, without being suffocated by the entropic residue of its present?

We propose that a key part of the answer is topological, not merely thermodynamic: a living system must maintain a connected transport field through which the boundary conditions governing its

own dynamics remain internally modifiable. Sections 3 through 6 situate this criterion in relation to prior definitions of the living state, develop it, relate it to Igamberdiev's recent re-evaluation of Ervin Bauer's principle of sustained non-equilibrium, and situate it in relation to recent anatomical evidence for body-wide interstitial continuity.

2. Boundary Conditions as Internally Modifiable Physical Parameters

In classical physics, a system is typically defined by boundary conditions imposed externally and held fixed: the walls of a container, the temperature of a reservoir, the geometry of a domain. Living systems violate this picture in a specific and physically consequential way. We define life as a physical state in which the junction between past structure and future possibility remains open through the continuous updateability of internal boundary conditions.

Here, boundary conditions are not treated as static external constraints but as internally modifiable physical parameters governing energy flow, transport pathways, and dissipative degrees of freedom. This updateability enables a biological system to remain globally non-equilibrated despite ongoing entropy production. By dynamically reshaping its own internal constraints, a living system avoids entrapment in local equilibrium states. Boundary-condition updateability thus functions as the physical bridge that allows structure inherited from the past to manifest as coherent activity in the future, without being suffocated by the entropic byproducts of the present.

This account does not depart from the companion paper; it asks what the interstitial gel's phase state -- the specific physical substrate examined there -- is an instance of. There, the swollen phase permitted continued modification of the thermal, osmotic, and ionic boundary conditions surrounding embedded cells, while the collapsed phase foreclosed it. Here, we treat that specific case as one instance of a more general physical criterion for the living state.

This criterion admits a more precise formulation in terms of Markovian dynamics. A Markov process is one in which the future state depends only on the present state, with no dependence on the trajectory by which that state was reached; the system, in this sense, has no memory. Living systems are characteristically history-dependent and may therefore require a non-Markovian description at appropriate coarse-grained scales: their present state carries the trace of the path taken to reach it -- in healed tissue, in immunological memory, in the mechanical history retained by extracellular matrix -- and their behaviour is shaped not only by where they are but by the range of states still reachable from there. We note that history dependence is not unique to living systems; glasses, viscoelastic materials,

and many other physical systems also require non-Markovian description. The distinction we draw is not the mere presence of memory but what accompanies it: in a glass, history is retained passively, as residual structure left by a trajectory the system can no longer influence, and the accessible future state space contracts monotonically toward the relaxed state. In a living system, history is retained while the accessible future state space is actively maintained -- and, crucially, can be enlarged, as in wound healing, immunological adaptation, or tissue remodelling. What we propose is therefore narrower than an appeal to non-Markovianity as such: that at an appropriate coarse-grained level, boundary-condition updateability can be viewed as the maintenance of a non-Markovian dynamical structure in which memory and future accessibility are jointly and actively sustained, rather than memory being a passive residue of an already-foreclosed trajectory.

On this formulation, the loss of updateability is Markovianization. As hysteretic memory is erased and the set of reachable states contracts, the system's dynamics collapse toward dependence on the present state alone. This is, we suggest, a more tractable statement of what is lost in chronic disease and in death than the language of energy flux permits: what fails is not the supply of energy but the system's capacity to carry its past forward and to keep its future open. We note, to forestall a natural objection, that this is not a claim that beginning from new conditions constitutes Markovianization. A plant transplanted to new soil begins from a new boundary condition while retaining its internal history and its capacity to respond to a new environment; it remains non-Markovian. Markovianization refers specifically to the loss of that capacity -- the erasure of history and the closure of possibility -- not to a change of circumstance.

A recent independent theoretical treatment has reached a structurally similar conclusion from a different starting point, formulating organismal aging and death within a non-Markovian stochastic thermodynamics framework and identifying the loss of dynamical stability, rather than the cessation of energy flux, as the physical signature of death [8]. That treatment develops the non-Markovian formalism in considerably greater mathematical depth than we attempt here, but remains at the level of an abstract state-space description; it does not specify what physical substrate, in any particular organism, carries the system's non-Markovian memory or constitutes the reachable future state space whose contraction it identifies with aging. This is precisely the question the interstitial framework developed here is intended to address, and we regard the convergence of two independent approaches on a non-Markovian characterization of the living state as suggestive rather than coincidental.

3. Relation to Prior Definitions of the Living State

The proposal that life consists in the maintenance of its own conditions is not new, and it is worth stating precisely what the present account adds. The closest antecedent is the theory of autopoiesis developed by Maturana and Varela, in which a living system is defined as one that continuously produces the boundary that defines it: the boundary generates the interior, and the interior in turn regenerates the boundary, in a closed loop of self-production [9]. Cessation of that self-production is, on their account, death.

The correspondence with boundary-condition updateability is close enough that the two may be read as the same insight expressed in different registers. Where autopoiesis speaks of a system producing its own boundary, we speak of a system modifying the boundary conditions governing its own dynamics; where autopoiesis identifies death with the cessation of self-production, we identify it with the loss of updateability. We regard this convergence as a point in favour of both accounts rather than a redundancy.

Two features of the subsequent literature make this correspondence more than incidental. The first concerns multicellularity. Maturana and Varela characterized multicellular organisms as second-order autopoietic systems -- systems composed of autopoietic units -- but explicitly declined to determine whether multicellular organisms are themselves first-order autopoietic systems, on the grounds that the molecular processes that would constitute them as autopoietic unities comparable to cells were unknown [10]. Weber and Varela subsequently argued that multicellular organisms are not autopoietic units in their own right. Razeto-Barry, reformulating the definition, reaches the opposite conclusion: on his reformulation, multicellular organisms are clearly first-order autopoietic systems [10]. Notably, he observes that while multicellular organisms do produce their own boundary -- skin in vertebrates, cuticle in plants -- the production of that boundary is not what physically maintains the system as a unit, in contrast to the unicellular case; what maintains multicellular cohesion, he suggests, are the chemical products that hold cells together. The question of what physically sustains a multicellular organism as a single spatially discrete unit is therefore, within this literature, explicitly open.

The second, and for present purposes more significant, concerns what autopoiesis does not specify. Razeto-Barry identifies as the principal shortcoming of autopoiesis as a definition of life that it is descriptive rather than mechanistic-causal: the property it names does not refer to causes or forces analyzable within physical, chemical, or thermodynamic theory, and in particular does not specify the thermodynamic conditions that allow the system to function as an autonomous machine, nor define its energetic relation to the environment [10]. This is not an external criticism imposed upon autopoiesis

from a rival framework; it is a limitation acknowledged from within the literature that defends the concept.

What we believe the present framework adds is a candidate answer to precisely this open question: what physically permits the boundary to be produced, and what thermodynamic conditions permit the organization to be sustained. Maturana and Varela's formulation is deliberately organizational rather than material -- it specifies the logical structure of self-production without specifying the physical substrate through which that structure is realized in any particular organism. The framework developed here proposes one such substrate for the multicellular case: the interstitial matrix in its swollen phase may provide candidate physical conditions -- diffusive, osmotic, mechanical -- under which the cellular boundary can continue to be produced and modified, and under which cells remain in the physical proximity that multicellular cohesion requires, while collapse of that matrix forecloses both. On this reading, the volume phase transition described in our companion paper is a candidate physical mechanism by which autopoiesis is locally suspended in tissue, and interstitial phase state is a candidate material determinant of whether autopoietic organization can be sustained.

4. The Dynamic Signature: Topological Connectivity Versus Passive Diffusion

The observable physical signature of boundary-condition updateability is sustained, non-diffusive persistence within the biological transport field. In the absence of active maintenance, transport fields generally tend toward diffusive relaxation and passive homogenization; we do not claim this tendency is unique to inert matter, since non-equilibrium self-organization -- convection, reaction-diffusion patterning, and dissipative structures more generally -- is well documented in inorganic systems as well. What distinguishes the living case, on our account, is not the mere presence of non-equilibrium organization, but its active, ongoing maintenance through internally modifiable boundary conditions: living systems preserve directed flow across scales within what we term the organism-wide mesoscopic transport field, an internal medium that remains dynamically connected rather than diffusively homogenized.

Such persistence requires nontrivial topological connectivity, characterized by a first Betti number $b_1 \geq 1$ -- ensuring the existence of continuous loops and alternative pathways for flow, rather than a tree-like or disconnected structure with no redundancy. We focus on b_1 specifically because it is the invariant that counts independent cycles: connectedness alone ($b_0 = 1$) permits a tree-like network with no redundancy at all, whereas $b_1 \geq 1$ guarantees at least one closed alternative pathway within the

network. We note that this is weaker than a guarantee of full fault-tolerance: a single loop attached to an otherwise tree-like structure still has $b_1 = 1$, yet a failure at the junction connecting that loop to the rest of the network can still disconnect it. $b_1 \geq 1$ is therefore best understood as a minimal, necessary structural signature of redundancy, not a sufficient guarantee against any single point of failure. Higher Betti numbers, which count higher-dimensional voids, may also be relevant to the three-dimensional geometry of the interstitial network, but we do not develop that possibility here. Topologically, b_1 -connectivity functions as an architectural radiator: it allows entropy to be distributed and dissipated more broadly across the network rather than trapped and accumulated at isolated nodes. This function is not confined to internal redistribution. Because the transport field is, as discussed below, continuous with the exchange the organism maintains with its external environment, the loss of connectivity we describe threatens not only the internal circulation of entropy but the field's openness to that external exchange, on which dissipative organization ultimately depends.

We emphasize that $b_1 \geq 1$ is proposed here as a candidate necessary structural signature of updateability, not as a sufficient criterion for life. Many inert physical networks -- a looped irrigation system, an electrical grid with redundant circuits -- also satisfy $b_1 \geq 1$ without being alive. What we propose is narrower: that among the further conditions required for a system to sustain boundary-condition updateability, nontrivial topological connectivity of its transport field is one necessary, measurable structural precondition. The full criterion for life, on this account, remains updateability itself; connectivity is a candidate physical substrate through which it becomes measurable, not a restatement of the definition.

This requirement is not a mathematical abstraction imposed upon biological tissue from outside. Recent anatomical work provides direct, independent empirical support for exactly this kind of connectivity in the human interstitium. Building on the initial characterization of the interstitium as a fluid-filled, body-wide connective tissue network [5], subsequent studies have demonstrated that interstitial spaces are continuous not merely within single organs but across organ and tissue boundaries: continuity has been shown between the pancreas and all adjoining tissues, along the adventitia of the entire cardiovascular system, and along the perineurium of the entire nervous system [3]. A parallel study demonstrated equivalent continuity of interstitial spaces within and beyond the human lung [4]. Together, these findings indicate that the interstitial network forms a single, anatomically continuous transport field spanning organ boundaries -- an architecture consistent with, though not a direct measurement of, the topologically nontrivial ($b_1 \geq 1$) connectivity that boundary-

condition updateability, as we define it, requires. The anatomical studies cited demonstrate spatial continuity of tissue; they do not themselves report a Betti number, and we do not claim they do.

An important distinction must be drawn here between geometric and functional connectivity. A tissue region may retain its anatomical architecture -- the channels remain structurally present, and a purely geometric analysis of the fixed tissue would still yield $b_1 \geq 1$ -- while the gel occupying those channels has collapsed to a state in which flux no longer passes. Geometrically the loop persists; functionally it does not. We therefore distinguish geometric topology, which describes the anatomical arrangement of the network, from effective topology, which describes the connectivity actually available to transport. It is effective topology that our framework concerns.

This distinction clarifies what we take the volume phase transition to accomplish. Gel collapse need not destroy the anatomical channel in order to abolish its function: by dehydrating and contracting the medium within the channel, the transition can reduce effective connectivity toward zero while leaving geometric connectivity intact. This is, we suggest, why the collapsed state is so difficult to detect by conventional structural imaging, which reports geometry rather than effective transport. Within this connected field, then, the reduction of effective b_1 toward zero in a region of tissue constitutes a local foreclosure of updateability: interstitial gel collapse does not merely dehydrate tissue, it functionally severs loops in the transport network, converting a connected, radiator-like architecture into an isolated, diffusion-limited pocket, without necessarily altering what an anatomical section would show.

A structurally analogous phenomenon has been independently demonstrated at a smaller spatial scale, within the cytoplasm of individual living cells. The crowded, glass-like cytoplasm is fluidized by mechanical energy derived from metabolism, converting a jammed, diffusion-limited intracellular medium into a more mobile, non-equilibrium state; this metabolically driven fluidization has been directly quantified through the violation of the fluctuation-dissipation theorem in living cells [6,7]. We do not claim that intracellular cytoplasmic fluidization and mesoscale interstitial VPT are the same phenomenon, nor that they share a common molecular mechanism; the spatial scales, constituent materials, and driving processes differ substantially. We note the parallel because it suggests that metabolically or mechanically driven transitions between a jammed, transport-limited state and a connected, transport-permissive state may be a recurring physical motif across the scales of living organization, from the single cell to the organism-wide interstitial field -- a motif for which boundary-condition updateability offers one candidate unifying description.

This raises the question of what plays the role of the transport field in organisms that have no interstitium at all. In unicellular organisms, the external environment itself serves this function: at the origin of life, that environment was the ocean. Multicellularity may on this view be understood as the internalization of the transport field -- the function once performed by the surrounding medium being taken over by an internal one. If this reading is correct, the interstitium is not a specialized tissue that happens to matter, but the particular form taken, in multicellular organisms, by a transport field that all life has required. This formulation is consistent with the thermodynamic truism that living systems require, by definition, exchange across their boundary: no system is open without an interface with something external to it. What differs across scales and modes of biological organization is what serves as the transport field, not whether one is necessary.

5. Convergence with Bauer's Principle and Igamberdiev's Thermodynamic Buffering

This account converges, from a different starting point, with a research program recently re-articulated by Igamberdiev [2]. Igamberdiev has argued that living systems are organized around a stable non-equilibrium state -- the principle first formulated by Ervin Bauer -- maintained through thermodynamic buffering: the coupling of near-equilibrium buffering reactions with far-from-equilibrium driving reactions, such that the system as a whole is stabilized in a regime of sustained non-equilibrium rather than relaxing toward thermodynamic death.

Boundary-condition updateability, as developed here, proposes a candidate physical substrate for how such buffering might be realized architecturally. Thermodynamic buffering requires that the system possess alternative pathways -- routes through which flux can be redistributed when one channel becomes obstructed or saturated. Topological connectivity ($b1 \geq 1$) is one plausible structural precondition for this kind of redistribution: a connected, loop-containing transport field provides an architecture through which buffering reactions could redistribute flux, whereas its absence would confine near-equilibrium and far-from-equilibrium components to a single, unbuffered pathway. We do not claim this connectivity is strictly necessary for thermodynamic buffering as Igamberdiev describes it, nor that thermodynamic buffering and topological connectivity are the same object: buffering is a description of reaction kinetics, and connectivity is a description of network geometry. The two frameworks may be complementary rather than mutually reducible; what we suggest is that they are compatible, and that the kind of connected transport field we describe here offers one structural setting in which the kinetic buffering Igamberdiev describes could plausibly operate.

This convergence is, we believe, mutually clarifying rather than redundant. Igamberdiev's framework specifies the dynamical and biochemical mechanism -- near-equilibrium enzymatic buffering -- by which sustained non-equilibrium is achieved at the metabolic scale. The present framework specifies a mesoscale architectural condition -- topological connectivity of the interstitial transport field -- that may provide the structural space in which such buffering mechanisms can operate. The two frameworks may therefore be complementary rather than mutually reducible; each operates at a scale the other does not directly address.

A further, more direct link is worth noting. The buffering reactions Igamberdiev describes are themselves enzymatic processes, and enzymatic function of this kind depends on the diffusive availability of substrates, ions, and water within the reaction milieu. The interstitial gel's swollen phase is precisely the physical condition under which such diffusion proceeds freely; the collapsed phase, by reducing water activity and constraining ionic and molecular mobility, may impair the same buffering chemistry Igamberdiev describes, independent of whether the relevant enzymes and substrates remain present. On this view, gel collapse does not merely sever topological loops in the transport network, as discussed above; it may also directly compromise the hydrated milieu on which near-equilibrium buffering chemistry itself depends. We offer this as a further candidate link between the two frameworks, not as an established mechanism.

6. Clinical Implications: Interventions as Topology Restoration

This framework reframes the goal of therapeutic intervention in cases of chronic disease. On the account developed in our companion paper, interventions such as interstitial hydromechanical modulation (IHM) -- the delivery of a localized osmotic, thermal, and mechanical perturbation into loose connective tissue spaces -- act by supplying the energy required to overcome the hysteresis barrier separating the collapsed and swollen phases of interstitial gel [1].

The present framework adds a topological interpretation of the same intervention: because the interstitial network is anatomically continuous across organ and tissue boundaries [3,4], a local restoration of the swollen phase at one anatomical site may, in principle, increase the effective connectivity of the network at that site and across the portion of the network to which it is topologically joined. We emphasize that what such an intervention would restore is effective rather than geometric connectivity: the anatomical channels were present throughout, and what changes is whether flux can pass through them. If this framework is correct, this change should in principle be reflected in an increase in measures of effective transport connectivity, though no such measurement has yet been

attempted, and we note that the appropriate measurement would need to assess transport rather than structure alone. This would be consistent with clinical observations, reported elsewhere, of therapeutic effects appearing at sites remote from the point of intervention and following vascular-anatomical rather than neural-segmental distribution [1]. On this account, such interventions are best understood not as delivering a local repair, but as reopening a dissipative stage that had been topologically severed -- restoring the connected loops through which entropy can once again be managed more broadly rather than locally.

Avascular tissue compartments offer a theoretically useful setting for future investigation of this account, precisely because direct vascular explanations for a tissue-maintenance effect are less readily available there than in perfused tissue. Articular cartilage is a well-characterized example: direct vascular supply to cartilage is absent, and chondrocyte nutrition is conventionally attributed to diffusion from synovial fluid and, to a limited extent, from subchondral bone, rather than to direct capillary perfusion within the cartilage itself. Any tissue-maintenance effect observed in such a compartment would necessarily occur independent of vascular supply, which is consistent with, though it would not by itself establish, the possibility that restored interstitial connectivity -- rather than restored blood flow -- can support tissue maintenance in diffusion-dependent compartments. We flag this as a candidate direction for future clinical investigation of the connectivity account proposed here, rather than as a claim supported by data reported in this essay.

7. Limitations

This framework carries limitations that parallel those of the companion paper from which it extends. First, the identification of $b_1 \geq 1$ as the relevant topological invariant is, at present, a theoretical proposal rather than a directly measured property of living tissue; no in vivo measurement of the interstitial network's Betti number has yet been performed. The distinction between geometric and effective connectivity introduced in Section 4 compounds this difficulty: measuring effective connectivity requires assessing transport, not merely resolving structure, and we are not aware of an established modality that does so at the relevant mesoscale across large tissue volumes. Structural imaging alone would report geometric connectivity and would therefore be expected to under-detect precisely the state we describe.

Second, the anatomical evidence we cite for interstitial continuity [3,4,5] demonstrates connectivity in ex vivo tissue sections under specific staining protocols; it does not, by itself, establish that this connectivity is dynamically maintained, actively regulated, or subject to the phase-transition-

like collapse we propose. The synthesis offered here -- linking anatomical continuity, topological connectivity, thermodynamic buffering, and volume phase transition -- is a theoretical integration, not an empirically demonstrated causal chain.

Third, the relationship between boundary-condition updateability as defined here and Bauer's principle as re-articulated by Igamberdiev is proposed as a structural convergence; we do not claim mathematical equivalence between the two formalisms, nor that either account is reducible to the other.

8. Conclusion

We have argued that the physical distinction between a living system and an inert one may involve a topological component that is not captured by thermodynamics alone: what must be maintained is not merely a flux of energy, but a connected transport field through which that flux can be redistributed, and through which the boundary conditions governing the system's own dynamics remain open to revision. This reframes the volume-phase-transition account of chronic disease developed in a companion paper as a specific instance of a more general criterion, gives that criterion a formulation in terms of non-Markovian dynamics, brings it into structural dialogue with both the theory of autopoiesis and Igamberdiev's re-evaluation of Bauer's principle, and grounds it in independent anatomical evidence for body-wide interstitial continuity that none of these accounts, on its own, had previously invoked. Where autopoiesis specifies the organizational form of self-production and Bauer's principle specifies the thermodynamic condition of its persistence, the present framework proposes a candidate material substrate through which both may be physically realized in multicellular tissue.

What remains open is precisely what a topological criterion should leave open: whether the first Betti number is the correct invariant, whether it can be measured in living tissue, and whether the connectivity we describe is actively maintained or merely structurally present. These are, in our view, tractable empirical questions rather than merely philosophical ones, and we hope this essay's chief contribution is to have posed them in a form that future imaging and biophysical work can address directly.

Declarations

Author Contributions:

T.H. developed the technique generalized in this essay as interstitial hydromechanical modulation (IHM) -- his own method, Regulation Of Osmosis in the Tissue Space (ROOTS) -- approximately 35 years ago, and the clinical observations that motivated the present theoretical development, including the vibration phenomenon, remote therapeutic effects, and the operator-independence of outcomes, draw on this long clinical practice. T.F. conducted the theoretical analysis, developed the topological and non-Markovian formulation presented here, and drafted the manuscript. Both authors reviewed and approved the final manuscript.

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The authors used Claude (Anthropic) to assist with manuscript preparation. The authors take full responsibility for all content, arguments, and conclusions.

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